Screening routines for aerosol- and trace-gas-affected infrared radiances

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1 Introduction

Progress in the NWP results in data assimilation systems producing ever smaller corrections to the background field. This implies ever greater need for accurate observations and screening processes to ensure that the assimilated data is representative of relevant meteorological features and nothing else. In the context of infrared (IR) radiance assimilation, the primary focus in observation screening has traditionally been in removing cloud-contaminated data. It is only recently that other potential sources of contamination, such as aerosol and trace gases, have become significant enough for the development of dedicated screening routines.

The European Centre for Medium-range Weather Forecasts (ECMWF) started rejecting aerosol-affected IR radiances in March 2016. Before that, aerosol contamination was only diagnosed from cloud-affected data population such that there were no aerosol-induced data rejections at all. From 2016 onwards, the aerosol detection module is fully independent of the cloud detection and it rejects substantial amounts of data in regions where Saharan dust is frequent (Letertre-Danczak, 2016). As demonstrated in Fig. 1, omitting the dedicated aerosol screening results in undesirable cooling in the model troposphere, and it also produces measurable degradation in medium-range forecast scores. The currently operational setup does not separate clear and aerosol-affected channels from each other, but instead rejects full spectra at locations where aerosol is found.

To deal with contamination from anomalous trace-gas concentrations, ECMWF introduced a new screening step in July 2017. This detection scheme was developed in response to the excessive amount of Hydrogen Cyanide (HCN), that occurred during the Indonesian wild fires in 2015 and had implications on the use of IR channels near

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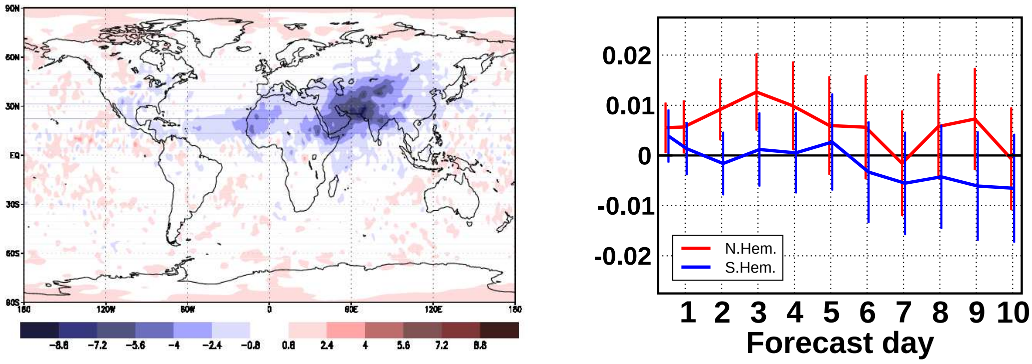

Figure 1: The impact of assimilating all channels in aerosol-affected IR FOVs on 500 hPa geopotential in 1 May–31 August 2019: (left) mean analysis difference (unit m^2s^{-2}). (right) Control-normalized forecast RMSE in the Northern (red) and Southern (blue) extratropics (positive values imply larger errors and bars indicate 95% confidence intervals).

the 712.5 cm^{-1} absorption line. The first implementation relies on brightness temperature (BT) observations only. In comparison with aerosol, the trace-gas contamination is less likely to cause problems in NWP, although it may occasionally produce spurious temperature and humidity analysis increments unless affected channels are removed.

Both the aerosol and trace-gas detection schemes have recently been revised for improved assimilation system performance. In the work on the aerosol detection, the goal has been to restrict the associated data rejections to the affected channels at each location, instead of the full-spectrum rejections at affected field-of-views (FOVs). The first version of channel-specific aerosol rejections is set for operational implementation in June 2020; an update including aerosol type classification will follow later. These new developments are described in Section 2. Also targeting a post-2020 operational implementation, the upgrade on the trace-gas detection provides added robustness and flexibility for further tuning by making combined use of observed BT and observation minus background (O-B) departures. This scheme is discussed in Section 3. The report is concluded by a short summary (Section 4).

2 The aerosol detection scheme

In the ECMWF scheme, aerosol is detected by looking for the characteristic "V-shape" (DeSouza-Machado et al., 2006) in observed BT spectrum across window channels both sides of O_3 absorption band at 1050 cm^{-1}. Specifically, positive detection of aerosol requires that

$$BT_{980} - BT_{1232} < t_1, \text{ and} \tag{1}$$
$$BT_{1090.5} - BT_{1234} < t_2, \tag{2}$$

where subscripts on the left-hand-side indicate channel wavenumbers (in cm^{-1}) and t_1 and t_2 are sensor-dependent tuning parameters. The chosen wavenumbers are consistent with Peyridieu (2010) and Peyridieu et al. (2010) and they are applied as such for IASI data. For AIRS and CrIS, these exact wavenumbers are unavailable, so nearest possible channels are used instead. To reduce the effect of noise, BT observation at each wavenumber is averaged over several channels. The tuning parameter values are chosen as a compromise between misses and false alarms, and MODIS satellite imagery of Saharan dust outbreaks are used as a reference here (Letertre-Danczak, 2016).

2.1 Aerosol type classification

For the purpose of making a distinction between affected and unaffected channels, it is helpful to first distinguish between different aerosol types. We have designed two additional tests with the aim at separating volcanic ash and Saharan dust from other aerosol species. The additional tests are only applied where some aerosol has already been detected. The first additional test exploits the concept of "ash BT difference" (Clarisse et al., 2010), and it classifies the detected aerosol as volcanic ash, if the inequality

$$BT_{1168} - BT_{1232} \quad < \quad t_3 \tag{3}$$

is true. Again, subscripts on the left indicate channel wavenumbers and t_3 is a tunable parameter. The test on volcanic ash is currently only applicable to IASI, as neither CrIS nor AIRS provides sufficient observation coverage at 1168 cm^{-1}. The second additional test, carried out only if the detected aerosol is not classified as volcanic ash, separates Saharan dust from other detected aerosol. The inequalities

$$BT_{1090.5} - BT_{833} \quad < \quad t_4, \text{ and} \tag{4}$$

$$BT_{1090.5} - BT_{1232} \quad < \quad t_5 \tag{5}$$

are required to be true for Saharan dust. FOVs that are found to contain some aerosol in the first test, but are classified as neither volcanic ash nor Saharan dust, remain currently unclassified in the ECMWF scheme.

Our experience suggests that this test produces realistic classification for Saharan dust and volcanic ash when applied to IR measurements over sea; this is illustrated in Fig. 2 for the case of Metop-B IASI data on 23 June 2019. However, for data over land surfaces, the performance is unconvincing. Therefore, our current practice is to continue rejecting full spectra at locations where aerosol is detected over land.

2.2 Aerosol-type dependent rejection thresholds

The way aerosol-affected channels are separated from unaffected ones depends on aerosol type. In each case, the separation makes use of channel height assignments (denoted h in the following). In the context of ECMWF NWP system, h are routinely calculated and used as part of the IR cloud detection scheme (McNally and Watts, 2003). The aerosol detection scheme identifies a scene-specific threshold height, against which each

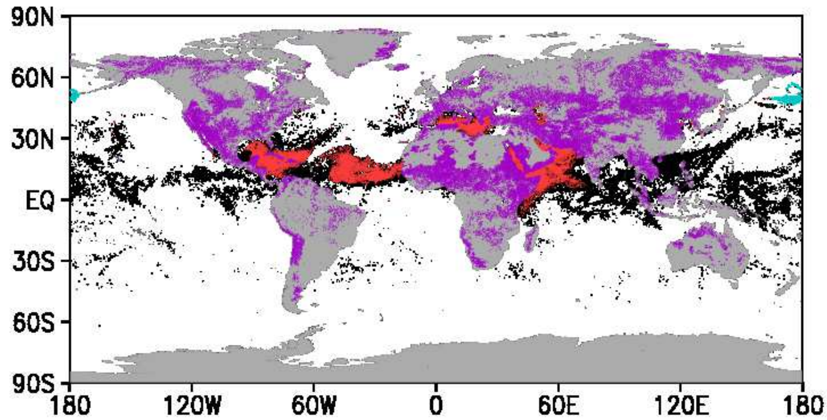

Figure 2: Aerosol type classification for Metop-B IASI on 23 June 2019. Red, light blue and black dots indicate presence of Saharan dust, volcanic ash, and unclassified aerosol, respectively. Purple dots indicate locations where aerosol is detected over land.

channel is compared. Only those channels, that fall below the threshold, are rejected: higher-assigned channels are considered free from aerosol contamination. For practical convenience, we use normalized the height assignments h_n, defined as

$$h_n = \frac{h - h_{min}}{h_{max} - h_{min}}, \tag{6}$$

where h_{min} and h_{max} are the minimum and maximum heights assigned at that particular FOV. Lower values of h correspond to higher-assigned channels, such that h_n=0 and h_n=1, respectively, represent the highest- and lowest-ranked channels.

Depending on the aerosol type classification, one of three possible routes is taken to determine the scene-specific threshold height h_{thres}:

1. The simplest route is to set h_{thres}=0 and thus reject all channels at locations where aerosol is detected. This route is chosen with volcanic ash and with any aerosol over land. On ash, it is noted that volcanic plumes can potentially reach stratosphere. However, such violent eruptions are not common, so it would take a long time to collect enough data for training the scheme. Over land, we choose the simple route due to the lack of confidence on the aerosol type classification.

2. On Saharan dust, we assume that Aerosol Optical Depth (AOD) is indicative of the top altitude of aerosol layer and that AOD can be approximated with a regression-based formula taking inter-channel BT differences as input. This way we can derive h_{thres} from BT observation. The procedure is described in detail in the next section.

3. For unclassified aerosol, the associated data rejections are limited to tropospheric-sensitive channels by setting h_{thres}=0.4. The fixed threshold is based on sensitivity experiments testing different values and evaluating overall NWP system performance in terms of background fit to independent observations.

2.3 AOD-dependent rejection on Saharan dust

For Saharan dust aerosol only, we make the rejection threshold dependent on AOD, which we estimate through the formula

$$AOD = \sum_{i=1}^{3} a_i (BT_{1090.5} - BT_{1234}). \tag{7}$$

Coefficients (a_i, i=1...3) are sensor-dependent and chosen such that the estimated AOD broadly corresponds to gridded AOD charts of the Copernicus Atmospheric Monitoring Service (CAMS). For dust radiative effect δ, we assume a linear AOD-dependence as

$$\delta = \alpha \, AOD, \tag{8}$$

where α is a channel-specific regression coefficient. Interpreting O-B departure as a proxy of δ, scatterplots shown in Fig. 3 give idea of the validity of the assumption made in Eq. (8), as well as of realistic values of α on two channels. The slopes of the least-squares-fitted lines indicate radiative effects of -0.7K and -2.6K from unit AOD of Saharan dust on these two channels. The plotted data includes all Saharan-dust-affected FOVs from the three Metop IASI's during 21–30 June 2019.

We have fitted the regression coefficients in a consistent way for all actively-used long-wave channels of IASI. The resulting slope coefficients are shown in Fig. 4 as a function of channel's mean normalized height assignment ($\overline{h_n}$). Indicating the strongest AOD-dependence, the lowest-ranked channels tend to show the largest (negative) slope coefficients. In the absence of any AOD-dependence, the slope coefficient is zero, as is the case on several high-peaking channels at the left-hand-side of the figure. In order for Saharan dust to have a measurable effect on O-B, channels need to be assigned with $\overline{h_n}$ >0.55. This is the range where slope coefficients are systematically non-zero.

Our next step is to parameterize the slope coefficient as a function of h_n. We do this by looking for a line that predicts α from given h_n, i.e.,

$$\alpha = \beta + \gamma \, h_n, \tag{9}$$

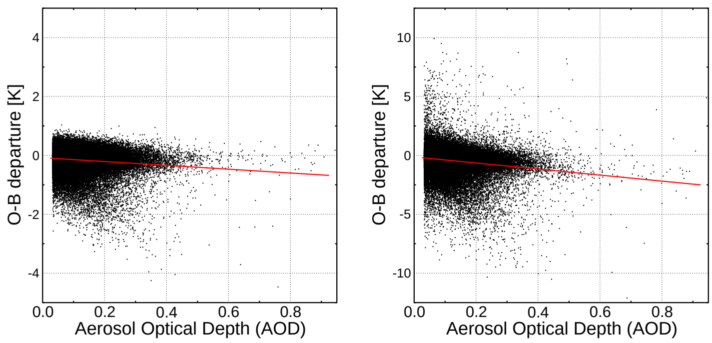

Figure 3: Linear regression for O-B departure as a function of estimated Saharan dust AOD on a mid-tropospheric (left) and lower-tropospheric (right) IASI channel.

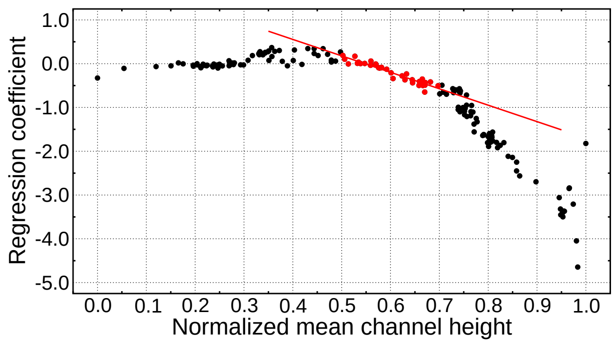

Figure 4: Slope of the (AOD, O-B) regression line as a function of IASI channel's mean normalized height assignment. The red line shows the least-squares fit through the slope coefficients when only accounting for data that falls between 0.5. . . 0.7 along the x-axis.

where β and γ are the line parameters. Judging from Fig. 4, the slope coefficients appear to follow one line in the x-axis range 0.5. . . 0.7 and another at 0.7. . . 1.0. We choose to constrain the line fitting in the former range (using the dots marked in red). This is somewhat subjective and motivated by our preference towars relatively conservative tuning, so that the fitted line (i.e., estimate of α) is on the negative side whenever h_n >0.55. The least-squares-fitted line parameters resulting from this choice are β=2.1K and γ=-3.9K.

Combining Eqs. (8) and (9), we derive the relation

$$h_n \quad = \quad \frac{1}{\gamma}\left[\frac{\delta}{AOD} - \beta\right], \tag{10}$$

that transforms into an AOD-dependent h_{thres} when δ is replaced by the desired maximum allowed dust radiative effect δ_{max}. The new aerosol detection scheme applies δ_{max}=-0.01K.

2.4 Evaluation

Performance of the new aerosol detection scheme is evaluated in terms of standard NWP system diagnostics taken from an experimental run where aerosol-related rejections are limited to affected channels only. In this context, control run rejects all channels at aerosol-affected FOVs. The experiment covers the four-month period 1 May–31 August 2019 and it is run at the horizontal resolution TCo399 using the version Cy46r1 of the Integrated Forecasting System (IFS).

As compared with the old scheme (control run), the new aerosol detection scheme lets more data through particularly on stratospheric-sensitive channels. The additional data concentrates over tropical Atlantic and Indian Oceans. On tropospheric-sensitive channels, there is relatively little extra data active in the experiment run. The meteorological impact of the added data is shown in Fig. 5. In comparison with the impact

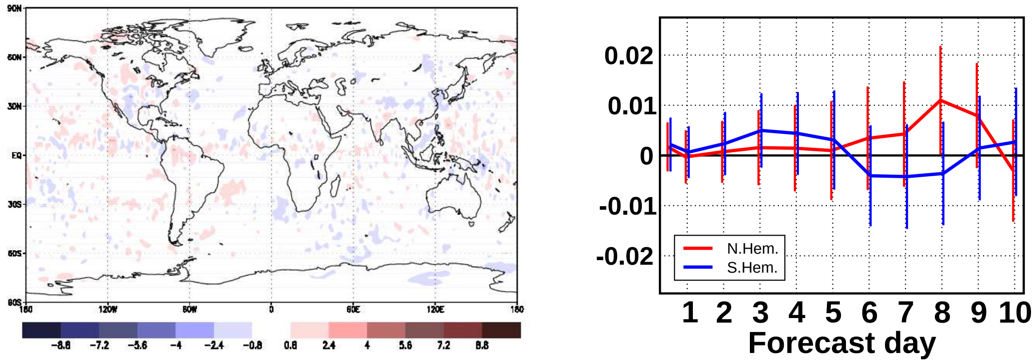

Figure 5: As Fig. 1, but showing the impact from using unaffected channels (rather than all channels) at aerosol-affected FOVs.

of not rejecting any aerosol-affected data (i.e., Fig. 1), the assimilation of unaffected IR channels does not show any particular impact on the mean analysis of 500 hPa geopotential height. Medium-range forecast scores indicate a neutral impact from the added data in both Northern and Southern extratropics, perhaps with the exception at forecast ranges beyond day-5 (where the impact is mixed).

3 The trace gas detection scheme

The trace gas detection scheme was developed soon after the 2015 episode of excessive HCN over the tropical Indian Ocean region. The episode manifested itself in the operational satellite data monitoring by showing persistent negative O-B departure on IR channels near 712.5 cm^{-1}. The spectral signature of the anomaly is a close match against the absorption spectrum of HCN. These aspects are illustrated in Fig. 6. A similar episode took place in September 2019, again in connection with considerable forest fire activity in Indonesia. Furthermore, time series of O-B data for AIRS in the ERA-Interim reanalysis shows indications of earlier occurrences in 2006 and 2010.

The ECMWF trace detection scheme relies on two-way comparison of observed BT and O-B departure amongst two channel groups. Channels measuring at the HCN absorption lines, called tracer channels, constitute one group, while the other group consists of control channels. Apart from the HCN sensitivity, the tracer and control channels are chosen such that their responses to temperature (and humidity) perturbations are similar. Within each group, the scheme calculates mean BT and mean O-B departure, and consequently produces tracer minus control differences for evaluation against pre-defined thresholds. The concept is demonstrated in Fig. 7. Straight lines indicate the thresholds set to isolate the contaminated data in the bottom-right corner of the plot. At the affected FOVs, rejection flags are raised for channels included in a pre-defined list.

Locations of HCN-affected CrIS FOVs are shown in Fig. 8 for two-day samples chosen at the peak time of 2015 (left) and 2019 (right) episodes. Separating cloud-

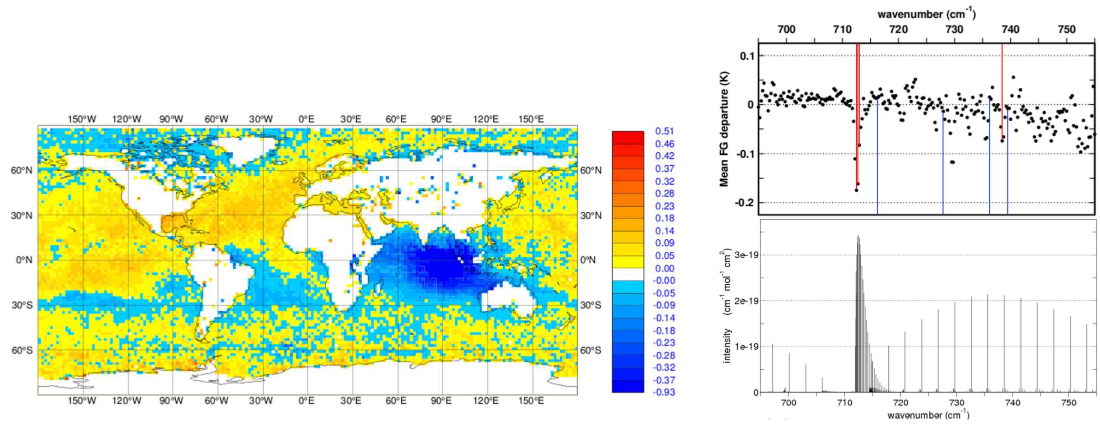

Figure 6: (left) Mean active O-B departure (in K units) at 712.5 cm^{-1} for NPP CrIS in the Autumn 2015. (right) Mean IASI O-B departure at the peak of the 2015 anomaly (top) against HCN absorption spectrum (bottom). Red and blue sticks indicate the tracer and control channels used in the detection scheme (see text).

affected (black) and cloud-free (red) locations (as flagged for the 712.5 cm^{-1} channel) demonstrates the importance of a specific screening routine for HCN. The middle panel represents a two-day period in October 2016, at a time of no significant concentration of atmospheric HCN. The few HCN detections in the 2016 sample are considered false alarms. Fortunately, most of these false alarms happen where the HCN-flagged channels would be rejected as cloud-contaminated anyway.

4 Summary

As part of the observation quality control in the ECMFW 4D-Var system, IR radiance observations are screened for contamination by clouds, aerosol, and trace gases. The importance of the screening is ever more critical as the 4D-Var system is correcting for ever smaller background errors and introducing ever smaller analysis increments. The screening routines for aerosol and trace gases are currently subject to refinement with associated changes in the operational system coming in Summer 2020 and at later upgrades. The new features in the aerosol detection scheme include (i) the separation of Saharan dust and volcanic ash from other aerosol species and (ii) limiting the associated data rejections to affected channels only. The trace gas detection is currently accounting for contamination by excessive HCN only. However, the implementation of the trace-gas detection is made flexible enough for easy extension to other gases as necessary.

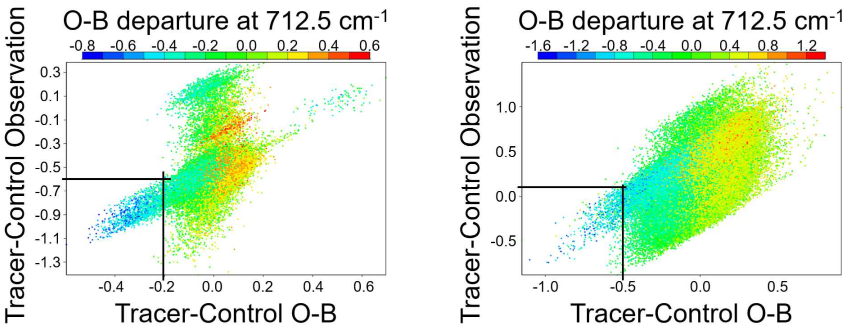

Figure 7: Scatterplots of tracer minus control mean BT observation vs. mean O-B departure in the 12Z analysis of 28 September 2019. Data includes S-NPP and NOAA-20 CrIS's (left) and Metop-A, Metop-B and Metop-C IASI's (right). Colour shading indicates O-B departure at 712.5 cm^{-1}.

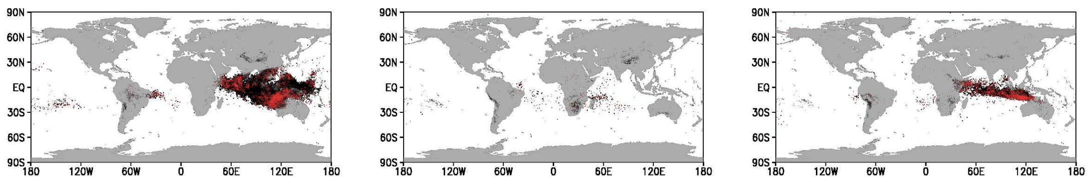

Figure 8: HCN-affected CrIS FOVs in two-day samples in 2015 (left), 2016 (middle), and 2019 (right). Red (black) indicates detection in the absence (presence) of cloud.

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